

PATNALA UDAY KUMAR

SOFTWARE ENGINEER | FULL-STACK & BACKEND DEVELOPER

udaypatnala5@gmail.com | +91 97036 60750 | [linkedin.com/in/patnala-uday-kumar](https://www.linkedin.com/in/patnala-uday-kumar) | github.com/UdayPatnala | patnala-uday-kumar.vercel.app

PROFESSIONAL SUMMARY

Software engineer focused on full-stack and backend development with hands-on experience building applications using Java, Python, JavaScript, React.js, and SQL. Designs REST APIs, Object-Oriented Programming (OOP) architectures, data layers, and responsive interfaces, with data manipulation using NumPy and Pandas, machine learning models, and software testing.

TECHNICAL SKILLS

Languages:	Java, Python, JavaScript, SQL
Core & Backend:	Java Backend, Object-Oriented Programming (OOP), Data Structures & Algorithms, DBMS, REST APIs
Web & Frontend:	React.js
Data Science & ML:	NumPy, Pandas, Machine Learning
Tools & Quality:	Git / GitHub, Software Testing, Debugging

PROJECTS

Music Mirror | React.js, Python, REST APIs *GitHub: github.com/UdayPatnala/music-mirror | Live Demo: music-mirror-aos.vercel.app*

- Developed a real-time webcam face-detection music recommendation system using React.js and Python API endpoints.
- Integrated embedded YouTube playback, manual mood overrides, and localStorage-backed profiles and mood history.

Churn Prediction System | Python, Scikit-learn, Machine Learning, Streamlit *GitHub: github.com/UdayPatnala/Churn-Prediction-System*

- Engineered an end-to-end ML pipeline with Scikit-learn utilizing hyperparameter optimization, scaling, and encoding for customer churn prediction.
- Developed REST API payload validation and an interactive Streamlit analytics dashboard for real-time risk assessment.

JavaPath Pro | React.js, Java, DBMS, SQL, REST APIs *GitHub: github.com/UdayPatnala/Java-Path | Live Demo: javapath-pro-aos.vercel.app*

- Built an interactive Java learning platform with real-time code execution, syntax diagnostics, and AI mentoring.
- Implemented password hashing, JWT authentication, protected REST APIs, and database persistence for user progress.

INTERNSHIP EXPERIENCE

Machine Learning Intern | Codec Technologies Pvt. Ltd. *Jun 2025 - Aug 2025*

- Developed data-preprocessing and machine-learning workflows using Python, NumPy, Pandas, and Scikit-learn during an 8-week internship.

Web Developer Intern | Codec Technologies Pvt. Ltd. *Jan 2026 - Feb 2026*

- Built and tested responsive web interfaces using HTML, CSS, JavaScript, DOM manipulation, and debugging tools during a 4-week internship.

EDUCATION

B.Tech in Computer Science Engineering (Data Science) | 2022 - 2026

Raghu Institute of Technology (Autonomous), Visakhapatnam, Andhra Pradesh | CGPA: 7.70/10

CERTIFICATIONS & ACHIEVEMENTS

- Programming in Java** - NPTEL (12 weeks, 2024)
- AWS Academy Cloud Foundations** (2024)
- Software Testing Masterclass** - Udemy Professional Certificate
- Published Research Paper:** "Smart Music Recommendation System Based on User Emotions," IJARESM, Vol. 14, Issue 3 (Mar 2026).